

Configurar un FTP implícito/explicito con SSL/TLS en Windows Server. Uso de sniffer



Clemente Cervantes Bustos

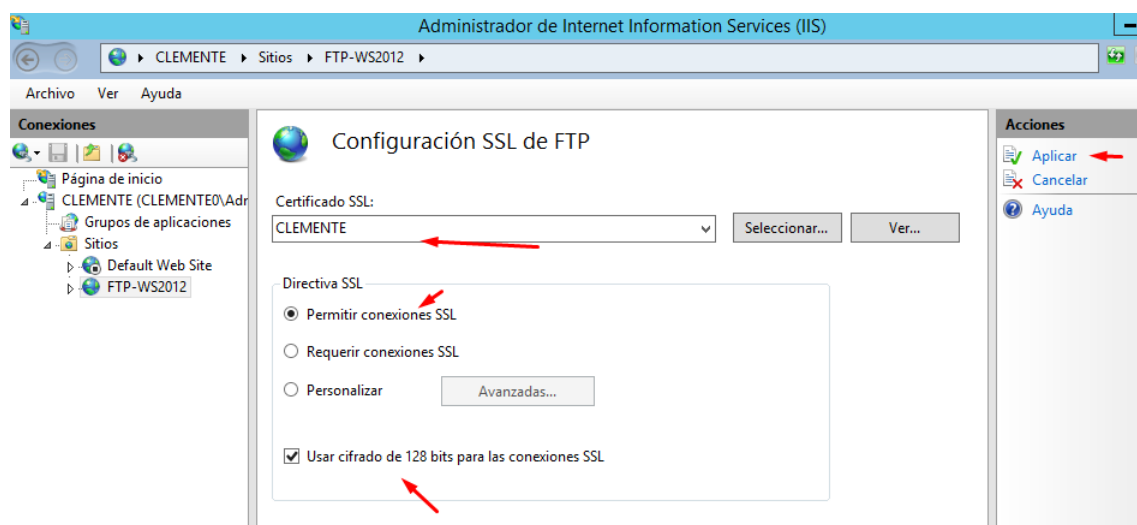
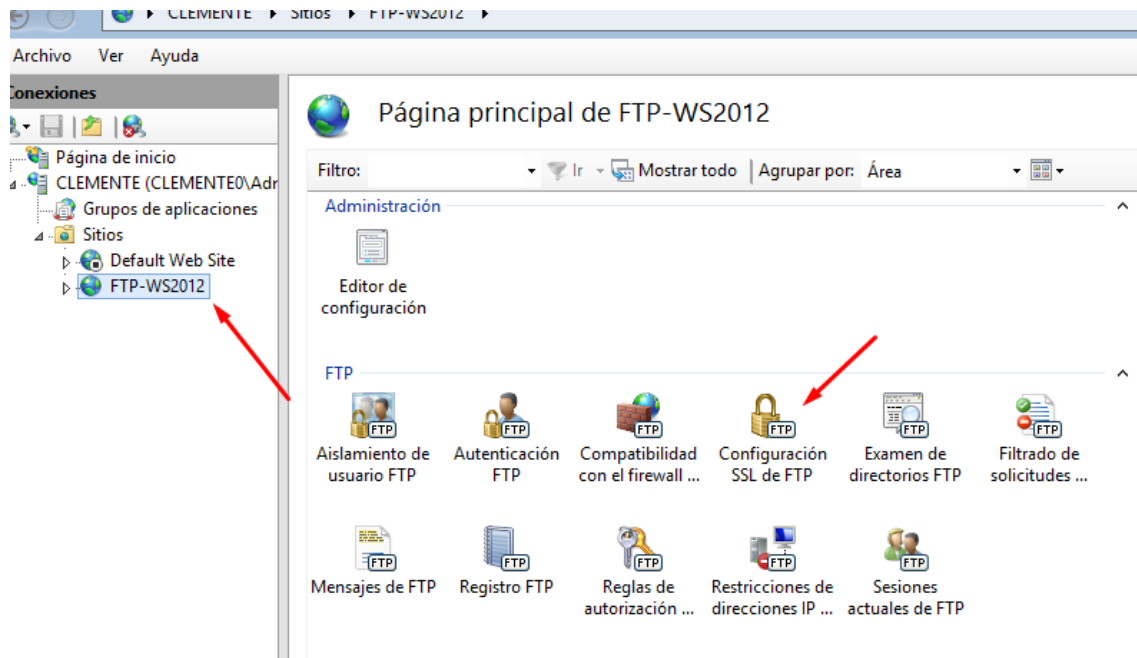
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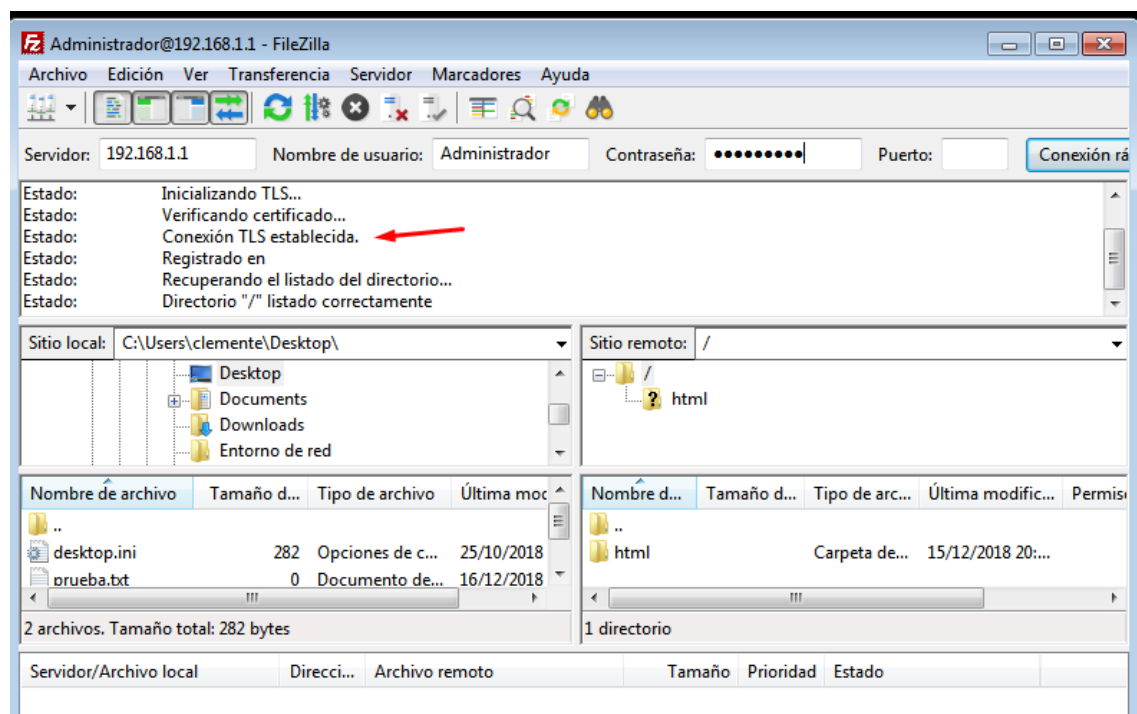
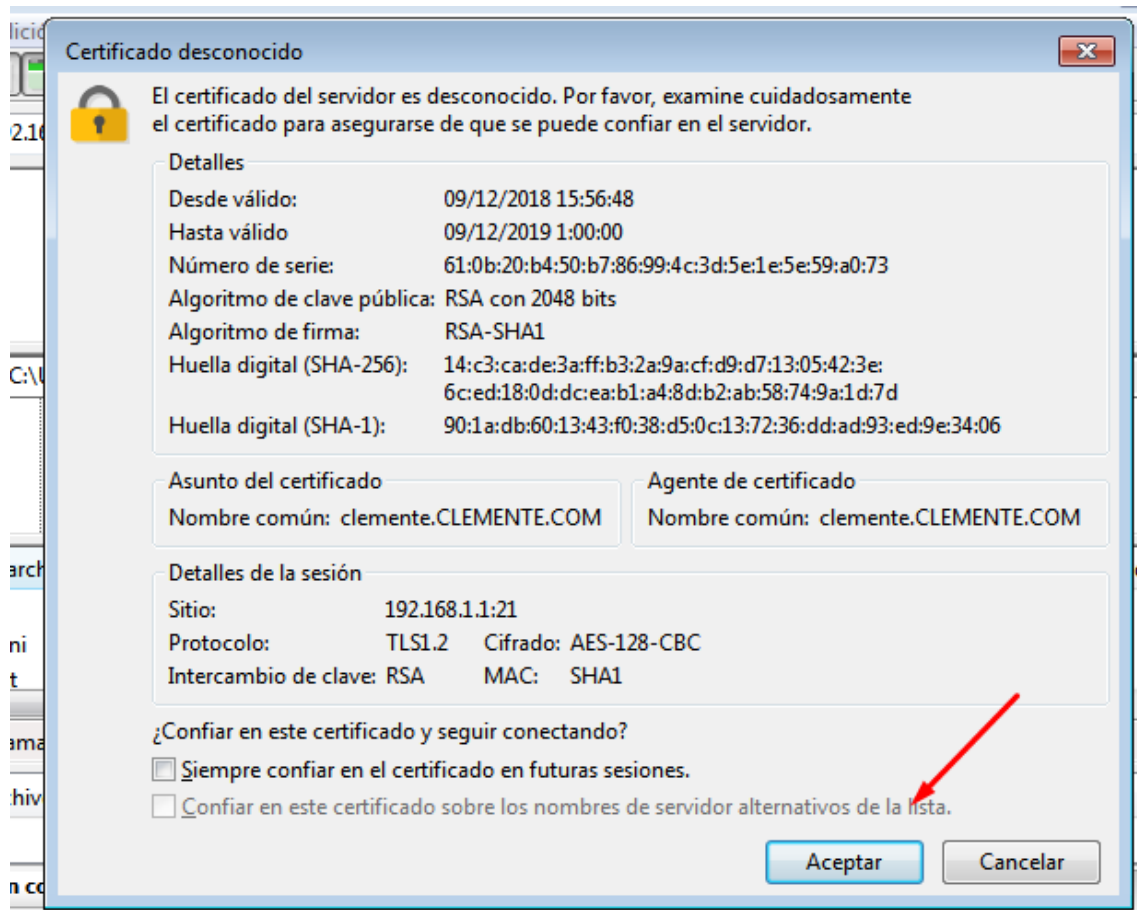
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1.- FTP explícito

Configuramos el SSL del FTP.



Desde Filezilla vemos como se conecta con SSL.



Podemos ver el tráfico como va totalmente cifrado.

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No.	Time	Source	Destination	Protocol	Length	Info
19	16.277699	192.168.1.1	192.168.1.100	FTP	81	Response: 220 Microsoft FTP Service
20	16.284532	192.168.1.100	192.168.1.1	FTP	64	Request: AUTH TLS ←
21	16.285548	192.168.1.1	192.168.1.100	FTP	103	Response: 234 AUTH command ok. Expecting TLS Negotiation.
22	16.300147	192.168.1.100	192.168.1.1	FTP	248	Request: 026(003)001(000)275(001)000(000)271(003)003(026)220(220)255w(276ik)(034)345(333):(230)021Y
23	16.301158	192.168.1.1	192.168.1.100	FTP	908	Response: 026(003)003(003Q)024(000)000(003)003(026)220(9)272:(222)212(326)255(317)272(003)241(017
24	16.303118	192.168.1.100	192.168.1.1	FTP	321	Request: 026(003)003(001)006(020)001(002)001(000R)3221(037)247~FQ(3340)334;(256j)314(0245)236
25	16.304087	192.168.1.100	192.168.1.1	FTP	68	Request: 024(003)003(000)001(001
27	16.305029	192.168.1.100	192.168.1.1	FTP	123	Request: 026(003)003(000)370(363)370,xu[24]265F(034)231)264(3326)2501(361)370(321)334)324(374)02
28	16.306038	192.168.1.1	192.168.1.100	FTP	129	Response: 024(003)003(000)001(001)026(003)003(000)337(377)177~(232)370M(333);306(234)362(314)002
30	17.061382	192.168.1.100	192.168.1.1	FTP	123	Request: 027(003)003(000)F243(003)327(h)223(222)237(272)023(326)322(257)271(242)6(210)320(342)2
31	17.063388	192.168.1.1	192.168.1.100	FTP	132	Response: 027(003)003(000)F243(003)327(h)223(222)237(272)023(326)322(257)271(242)6(210)320(342)2

The screenshot shows a Wireshark interface with a packet capture from a Windows Server-1 Ethernet0 to Ethernet0 switch-1 Ethernet0. The packet list on the left shows several FTP-related packets. Packet 22 is selected, which is an FTP Request (2026) from 192.168.1.100 to 192.168.1.1.

The details pane for packet 22 shows the following layers:

- > Frame 22: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits) on interface 0
- > Ethernet II, Src: PcsCompu_34:5e:1f (08:00:27:34:5e:1f), Dst: PcsCompu_e6:1e:05 (08:00:27:e6:1e:05)
- > Internet Protocol Version 4, Src: 192.168.1.100, Dst: 192.168.1.1
- > Transmission Control Protocol, Src Port: 49160, Dst Port: 21, Seq: 11, Ack: 77, Len: 194
- > File Transfer Protocol (FTP)

The hex and ASCII panes at the bottom show the raw data of the selected packet. A red arrow points to the ASCII representation of the packet data, which starts with '..'. ^4...E..

Si nos conectamos desde el terminal también nos podemos conectar, pero en este caso sin SSL.

```
C:\Windows\system32\cmd.exe - ftp 192.168.1.1
Microsoft Windows [Versión 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. Reservados todos los derechos.

C:\Users\clemente>ftp 192.168.1.1
Conectado a 192.168.1.1.
220 Microsoft FTP Service
Usuario (192.168.1.1:(none)): Administrador
331 Password required
Contraseña:
230 User logged in.
ftp>
```

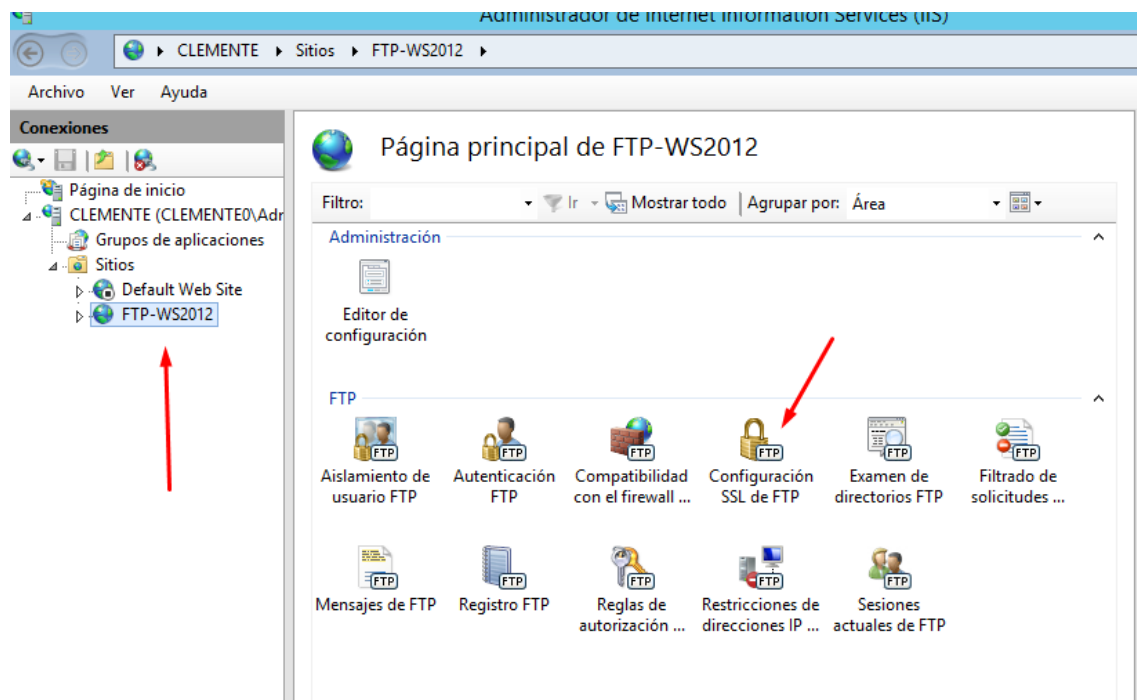
Podemos ver como el trafico va en texto plano.

*Standard input [Windows-Server-1 Ethernet0 to Ethernetswitch-1 Ethernet0]

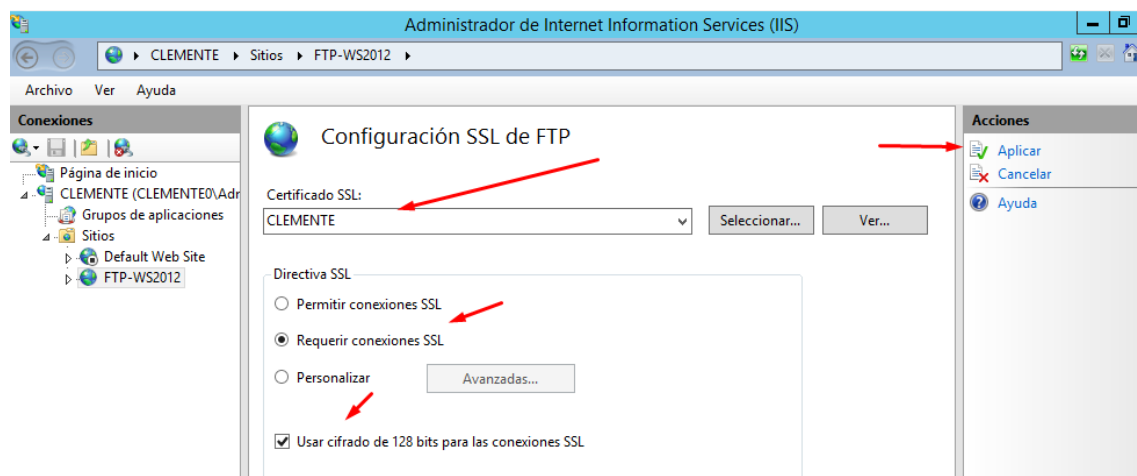
No.	Time	Source	Destination	Protocol	Length	Info
54	17.991638	192.168.1.100	192.168.1.1	FTP	107	Request: \027\003\003\000\266\360\333\035\335\376\206\224p\317 ;\271\362\235;\33
55	17.992614	192.168.1.1	192.168.1.100	FTP	155	Response: \027\003\003\000';\315E\236\247T\254e\243\230\371\305\310\332
56	17.994565	192.168.1.100	192.168.1.1	FTP	107	Request: \027\003\003\000\273;\372\264\255\340\267r@\216'QVa\223\352\352\202\33
57	17.995541	192.168.1.1	192.168.1.100	FTP	139	Response: \027\003\003\000P\273\336\#8c\274\320\230\253\004\364B\330\000m\037m\30
70	18.007253	192.168.1.1	192.168.1.100	FTP	123	Request: \027\003\003\000@\004\272\312\276\326\200\312\210q0+Z\341\324\353RK\37
122	117.889753	192.168.1.1	192.168.1.100	FTP	81	Response: 220 Microsoft FTP Service
125	120.753062	192.168.1.100	192.168.1.1	FTP	74	Request: USER Administrador
126	120.754966	192.168.1.1	192.168.1.100	FTP	77	Response: 331 Password required
129	123.954640	192.168.1.100	192.168.1.1	FTP	70	Request: PASS Inves2018
130	123.956592	192.168.1.1	192.168.1.100	FTP	75	Response: 230 User logged in.

2.- FTP implícito

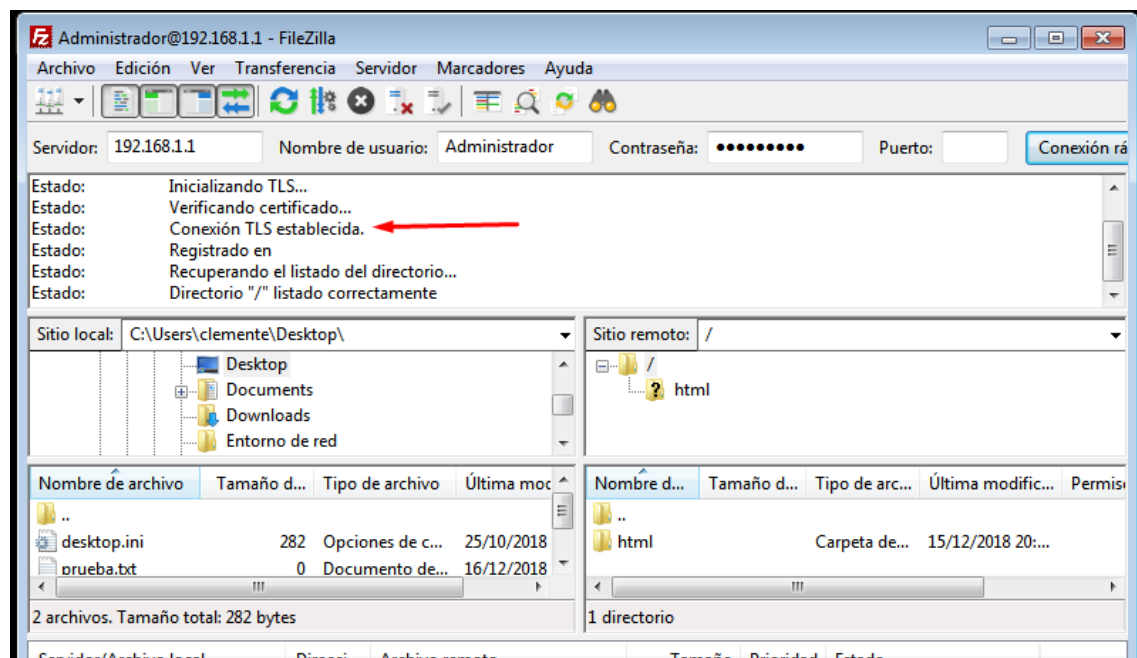
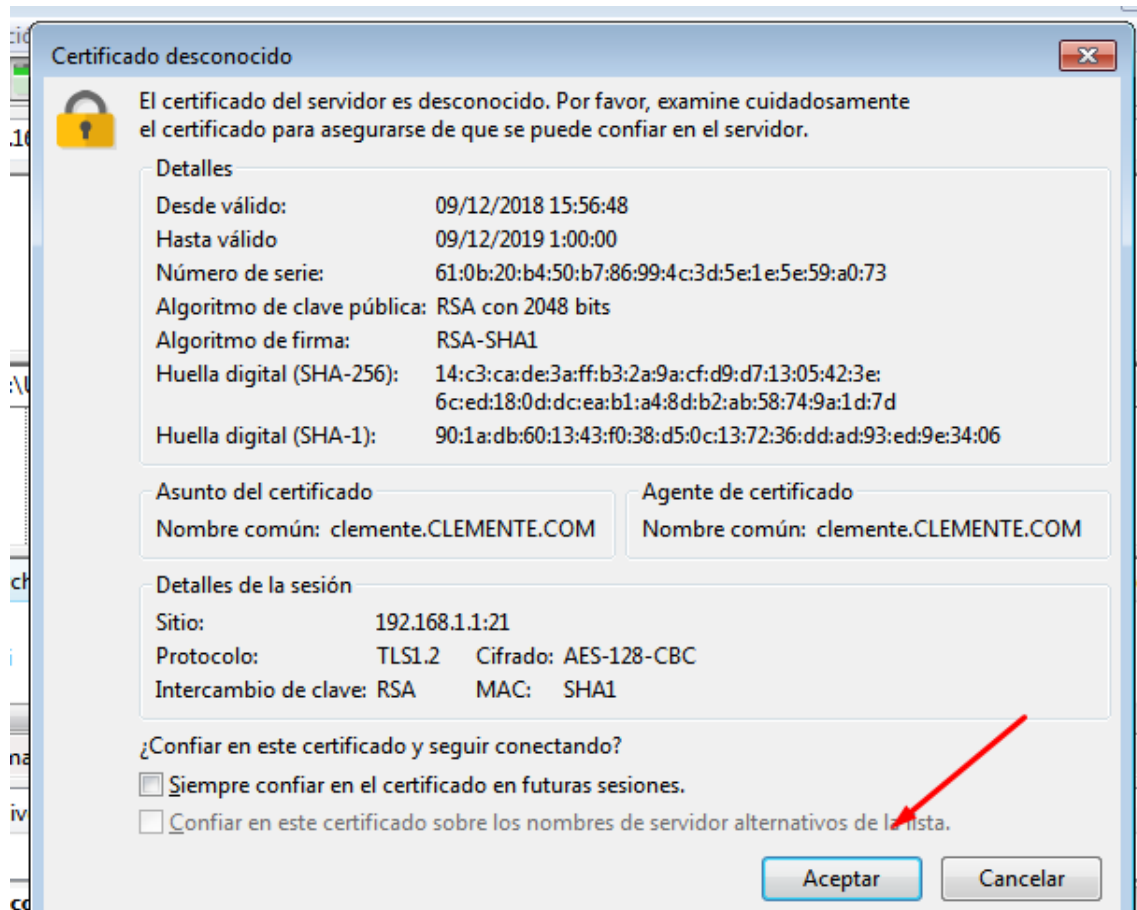
Nos vamos al mismo sitio de antes.



Pero esta vez elegimos la opción de requerir conexiones SSL.



La conexión en este caso con Filezilla es igual en ambos casos.



Pero en este caso si intentamos una conexión sin SSL no nos permite conectarnos.

```
C:\Windows\system32\cmd.exe - ftp 192.168.1.1
Microsoft Windows [Versión 6.1.7600]
Copyright (c) 2009 Microsoft Corporation. Reservados todos los derechos.

C:\Users\clemente>ftp 192.168.1.1
Conectado a 192.168.1.1.
220 Microsoft FTP Service
Usuario (192.168.1.1:(none)): Administrador
534 Policy requires SSL.
Error al iniciar la sesión.
ftp> _
```

Vemos el trafico como nos indica que la política requiere SSL.

*Standard input [Windows-Server-1 Ethernet0 to Ethernetswitch-1 Ethernet0]

No.	Time	Source	Destination	Protocol	Length	Info
197	388.784719	192.168.1.100	192.168.1.1	FTP	107	Request: \027\003\003\0000\352>\212\242v\311\257\247\177
198	388.785695	192.168.1.1	192.168.1.100	FTP	123	Response: \027\003\003\0000\257\234\313K
199	388.786637	192.168.1.100	192.168.1.1	FTP	107	Request: \027\003\003\00005\257e[\225jz\352V\006\2701J\2
200	388.787612	192.168.1.1	192.168.1.100	FTP	155	Response: \027\003\003\000`\231\251\210\225\2558\277\202
201	388.790588	192.168.1.100	192.168.1.1	FTP	107	Request: \027\003\003\0000\334\337}!M\003g\v
202	388.791515	192.168.1.1	192.168.1.100	FTP	139	Response: \027\003\003\000PQ\323tG\227\302\361 \346\360#
216	388.805181	192.168.1.1	192.168.1.100	FTP	123	Response: \027\003\003\0000@\371\253\237:\206\314QQ>\300M
238	436.346130	192.168.1.1	192.168.1.100	FTP	81	Response: 220 Microsoft FTP Service
240	438.919602	192.168.1.100	192.168.1.1	FTP	74	Request: USER Administrador
241	438.920578	192.168.1.1	192.168.1.100	FTP	80	Response: 534 Policy requires SSL.